

RUSHYANTH NERELLAKUNTA

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SUMMARY

AI/ML Engineer with 4+ years of experience designing and deploying production-grade GenAI systems, RAG pipelines, and multi-agent frameworks. Proven expertise in LLM orchestration, fine-tuning (LoRA/PEFT), and agentic architecture using LangGraph, Google ADK, and MCP. Demonstrated ability to deliver measurable impacts including a 40% reduction in response latency, 35% decrease in LLM API costs, and 95% accuracy on complex financial document queries. Skilled in end-to-end MLOps across AWS (Lambda, SageMaker, Bedrock), Snowflake, Docker, and CI/CD pipelines. AWS Certified Machine Learning Engineer, with an MS in Applied Data Science (GPA: 3.8) from IUPUI.

PROFESSIONAL EXPERIENCE

Software Engineer II (AI/ML) | Kahana Inc — Indianapolis, IN (Remote)

Aug 2024 – Present

- Designed a multi-agent system using Google ADK with specialized sub-agents handling routing and multi-step task execution (shopping, job apply, search). Standardized tool connectivity via MCP, accelerating tool-onboarding velocity by 60% across the platform's integration layer.
- Reduced response latency 40% by architecting a serverless inference pipeline on AWS Lambda, integrating Deepgram STT and Gemini APIs(intent detection) for a voice-first browser product serving real-time user traffic.
- Cut LLM API costs by 35% through semantic token caching and request-level prompt routing, reducing redundant computation across high-traffic voice interactions. Mitigated hallucination risk by implementing ReAct-style planning and guardrails with validation for voice context.
- Engineered a Supabase → Snowflake migration pipeline, re-architecting schema for columnar storage, query efficiency, and downstream analytics. Established CI/CD with GitHub Actions for reproducible builds, automated testing, and reliable ML service deployments.

Software Engineer (ML Research) | COMET Lab, Indiana University Indianapolis

Aug 2024 – May 2025

- Improved NL-to-SQL execution accuracy from 78% to 92% by implementing schema-aware RAG few-shot and CoT/ ReAct frameworks.
- Shipped a 60+ ground-truth evaluation harness with automated performance tracking, accelerating research iteration cycles for the lab.
- Enforced 100% safe execution via a read-only guardrails layer and Human-in-the-Loop review workflow for database query validation.

Software Engineer (ML Platform) | Virgenverse — Hyderabad, India

Feb 2022 – Nov 2023

- Reduced data pipeline refresh latency by 40% by automating ETL workflows with Python and Apache Airflow, increasing reliability across production data feeds. Boosted BI query performance 3× by migrating 10+ legacy stored procedures to optimized Snowflake transformations.
- Improved content classification accuracy by 18% using NLP (spaCy, NLTK) on a corpus of 100K+ social posts powering real-time marketing dashboards. Ran 6+ A/B experiments using statistical modeling and hypothesis testing, increasing VR app retention by 12%.

PROJECTS

Agentic RAG over SEC 10-K Filings • LangChain • ChromaDB • BGE • PyMuPDF • Python • [Link](#)

Jan 2026

- Processed 3,751 financial documents (2,016 pages + 1,735 tables) into a hierarchical chunking system (6,068 parent + 20,300 child chunks), reducing context fragmentation by 40% and achieving 95% accuracy on complex multi-year financial queries.
- Implemented RAG Fusion with Reciprocal Rank Fusion and a bi-encoder + cross-encoder two-stage retrieval pipeline, improving recall by 35% and cutting irrelevant results by 50%.

Medical QA- LLM Fine-Tuning Pipeline • HuggingFace • Pytorch • Unsloth • Microsoft phi3 • Lora • [Link](#)

Jan 2026

- Fine-tuned Microsoft Phi-3 Mini (3.8B) on 5,000 medical Q&A samples using 4-bit QLoRA (Unsloth), training only 1.47% of parameters (29.9M) on a single T4 GPU in under 90 minutes demonstrating parameter-efficient fine-tuning at scale
- Achieved +23% ROUGE-1, +126% ROUGE-2, +42% ROUGE-L, and BERTScore F1 gain from 0.826 → 0.848 vs. base model across 50 held-out test samples, with all evaluation fully automated

Hybrid Recommendation System • Python • PyTorch • FAISS • LightGBM • Kafka • Kubernetes • Redis • [Link](#)

Mar 2026

- Built a production-grade two-stage retrieve-then-rank pipeline: FAISS ANN retrieval (50M+ vector scale) → LightGBM re-ranking, with cold-start fallbacks, MMR diversity controls, and IPS-weighted bias mitigation serving personalized recommendations under 100ms.
- Designed end-to-end MLOps infrastructure: Prefect-orchestrated training (Spark ALS, Two-Tower, GRU), Feast feature store, MLflow experiment tracking, Evidently drift detection, and Kubernetes deployment with Prometheus/Grafana observability.

Real-Time Object Detection System • YOLOv8 • Kafka • Kubernetes • FastAPI • MLflow • [Link](#)

Mar 2026

- Reduced end-to-end pipeline latency from ~1,855,000ms (backlogged queue) to ~282ms — a 6,575× improvement — by redesigning Kafka consumer offset management and deploying on Kubernetes with Horizontal Pod Autoscaler (1→3 pods at 70% CPU).
- Trained YOLOv8s on COCO128 achieving 78.9% mAP50 at ~115ms inference latency; containerized FastAPI inference service with Docker for repeatable, scalable deployment.

EDUCATION

Master of Science, Applied Data Science | Indiana University Purdue University Indianapolis • GPA 3.8/4.0

May 2025

Bachelor of Technology, Electronics & Communication Engineering | Sreenidhi Institute of Science and Technology • GPA 3.2/4.0

May 2022

TECHNICAL SKILLS

- Languages:** Python (Expert), C++, SQL, JavaScript/TypeScript, Bash, R, MATLAB, Git, Github
- ML & Information Retrieval:** PyTorch, TensorFlow, JAX, FAISS, LightGBM, XGBoost, scikit-learn, RAG, LangChain/LangGraph, Hugging Face, spaCy, NLTK, LLM Fine-tuning (LoRA/PEFT), ChromaDB, Pinecone, Mathematics, Statistics
- Cloud & MLOps:** AWS (Lambda, SageMaker, S3, Bedrock, CloudWatch), Snowflake, MLflow, MongoDB, Evidently, Prometheus, Grafana, Google ADK, Model Context Protocol (MCP). **Certification:** [AWS Certified Machine Learning Engineer](#) (Jul 2025 – Jul 2028)
- Distributed Systems & Infrastructure:** Apache Kafka, Kubernetes (K8s), Docker, Apache Spark (PySpark), Redis, FastAPI, CI/CD (GitHub Actions), Airflow. **Other:** Data Structures, Algorithms, Agile Project Management, Huggingface, n8n, Tableau, PowerBI, Streamlit, MixPanel